

DSPO: STABLE AND EFFICIENT POLICY OPTIMIZATION FOR AGENTIC SEARCH AND REASONING

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ABSTRACT

Equipping LLMs with the capacity to query external knowledge sources is essential for tasks whose solution requires information beyond parametric memory. Existing approaches either depend on prompting to elicit latent tool-use behavior or exhibit performance saturation and policy collapse when RL is applied to complex interactive environments, thereby leaving substantial agentic capability unoptimized. We introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an RL algorithm that improves agent training by combining sequence-level policy optimization with dynamic sample filtering. The resulting model is trained solely from RL feedback to interleave multi-turn search with reasoning, eliminating the requirement for supervised demonstration trajectories. Across multiple QA benchmarks, our 7B model improves over comparable prior work by **34.1%**; on complex multihop QA such as HotpotQA, it also exceeds the corresponding 14B prior model by nearly **9% relative** while maintaining stable optimization throughout training.

1 INTRODUCTION

Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Zhao et al., 2023) achieve strong performance on specialized tasks, including mathematical reasoning (Shao et al., 2024; Trinh et al., 2024; Yu et al., 2025), code generation (Zheng et al., 2023; Yang et al., 2024), and creative writing (Chakrabarty et al., 2024; Marco et al., 2024). Nevertheless, their accessible knowledge remains bounded by the training distribution and is therefore static at inference time. A standard remedy is to augment the model with a search interface, yielding an agent that can issue external queries and condition subsequent generation on retrieved evidence (Jin et al., 2025b). This capability instantiates tool-calling (Schick et al., 2023): the policy must decide when to invoke a search tool, how to formulate the query, how to integrate retrieved content, and when to terminate with an answer. Learning this behavior requires optimizing a multi-step policy and is naturally formalized as a sequential decision-making problem suitable for Reinforcement Learning (RL).

In contrast to Supervised Fine-Tuning (SFT), which depends on expensive fixed demonstrations and does not directly optimize exploratory interaction, RL permits LLM policies to improve from trial-and-error feedback (Chu et al., 2025). Value-free algorithms such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) have therefore become attractive because they avoid an explicit value model and reduce memory requirements. However, when GRPO is transferred from constrained generation settings to open-ended interactive search, its optimization dynamics become unstable (Jin et al., 2025b; Yu et al., 2025; Cui et al., 2025; Liu et al., 2025). Two mechanisms account for this failure mode. First, as noted by Zheng et al. (2025), GRPO applies token-level likelihood ratios while the reward is assigned at the sequence level, producing high-variance gradient estimates. Second, sparse terminal rewards in search tasks frequently produce rollout groups with homogeneous outcomes, such as all-correct or all-incorrect samples; in such groups, the normalized advantage degenerates and the update contains little or no usable learning signal (Yu et al., 2025; Liu et al., 2025).

We address these coupled sources of instability and sample inefficiency with **Dynamic-filter Sequence-level Policy Optimization (DSPO)**. The algorithm combines two policy-optimization design principles in a single training procedure. First, DSPO adopts the sequence-level optimization principle of GSPO (Zheng et al., 2025), thereby matching the optimization unit to the reward unit and reducing variance in long-horizon reasoning trajectories. Second, DSPO incorporates a dynamic outcome-based filtering mechanism, inspired by DAPO (Yu et al., 2025), that admits only rollout groups containing both successful and unsuccessful samples for a prompt. This sampling criterion ensures that the group-relative advantage $\hat{A}_i$ is defined by non-zero reward variance and supplies an informative update. Integrated together, these components define a stable RL algorithm for multi-turn search-and-reasoning agents. Empirically, our

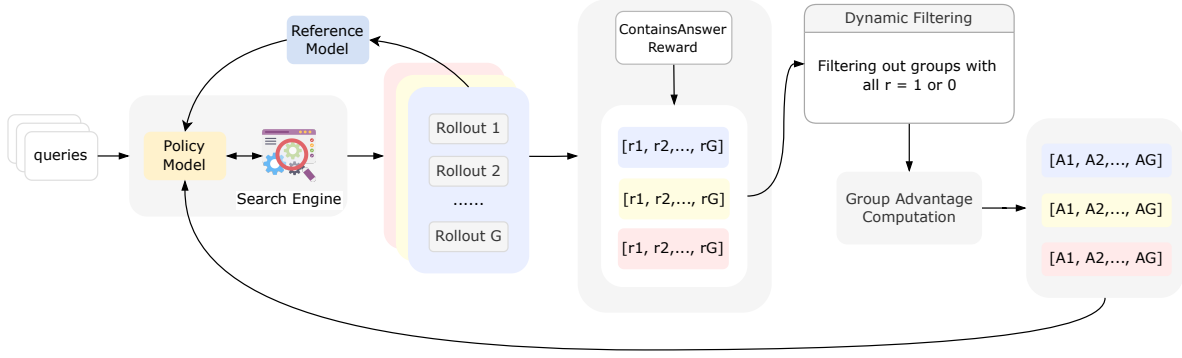


Figure 1: Overview of the DSPO training loop. For a query, the policy generates a group of G trajectories through interaction with the search environment, and each trajectory receives a sparse terminal reward. The **dynamic filter** removes groups with homogeneous outcomes and continues sampling until the batch is filled, so each retained batch contains an informative advantage signal. The resulting advantages are then used to update the policy under a sequence-level objective.

model obtains a **34.1% relative improvement** over a leading 7B baseline (Jin et al., 2025b); on HotpotQA, it also exceeds the 14B counterpart (Jin et al., 2025a) by **nearly 9% relative** (0.613 vs. 0.563).

The main contributions are as follows:

- We propose DSPO, an RL algorithm for agentic search that directly targets optimization instability and sample inefficiency. The method unifies **sequence-level optimization**, which stabilizes policy updates, with **dynamic outcome-based filtering**, which maintains an informative advantage signal under sparse rewards.
- We quantify DSPO’s performance gains through controlled benchmarking. The 7B DSPO model achieves a **34.1% relative improvement** over a comparable 7B baseline and surpasses the corresponding 14B model on complex multi-hop QA, with a nearly **9% relative gain** on HotpotQA (0.613 vs. 0.563).
- We provide empirical evidence that DSPO yields stable learning trajectories. Because all reported RL experiments use a basic BM25 retriever, the observed gains are attributable to the policy-optimization procedure rather than to a stronger retrieval backend.

2 RELATED WORK

2.1 RL FOR LLMs

RL methods for LLMs have progressed from Reinforcement Learning from Human Feedback (RLHF) pipelines that combine PPO with explicit reward models (Ouyang et al., 2022; Christiano et al., 2017; Schulman et al., 2017) toward more direct optimization procedures such as DPO (Rafailov et al., 2023). Group Relative Policy Optimization (GRPO) (Shao et al., 2024) represents a further shift toward value-free optimization by estimating relative advantages from statistics over response groups. However, the token-level GRPO objective can induce unstable training dynamics (Liu et al., 2025; Cui et al., 2025), motivating several refinements. GSPO replaces token-level ratios with a sequence-level objective aligned with sequence-level rewards (Zheng et al., 2025); DAPO improves sample efficiency under sparse rewards through dynamic outcome-based sampling (Yu et al., 2025); and GMPO stabilizes token-level updates through geometric-mean aggregation that reduces sensitivity to outlying ratios (Zhao et al., 2025). In our experiments, these methods still exhibit either collapse or performance bottlenecks in autonomous search settings. DSPO therefore synthesizes sequence-level optimization with dynamic filtering and filling in a unified algorithm tailored to the optimization pathologies of search-agent training.

2.2 LLMs WITH AGENTIC RETRIEVAL

To mitigate the static knowledge constraint of LLMs, retrieval-augmented generation (RAG) couples generation with external retrieval modules that supply time-varying or non-parametric information (Lewis et al., 2020; Gao et al., 2023). Classical RAG systems use dense retrievers to select relevant documents and concatenate the retrieved text

into the model context before generation (Karpukhin et al., 2020). Such pipelines are typically fixed, which limits autonomous control in complex multi-turn settings. Recent work instead treats LLMs as agents that iteratively plan, search, and reason. ReAct integrates reasoning traces with tool actions for tasks such as web navigation (Yao et al., 2023), and multi-agent frameworks such as AutoGen coordinate collaborative workflows (Wu et al., 2024). Additional agentic RAG approaches combine external tools with learned decision-making; for example, Wu et al. (2025) introduce Agentic Reasoning to streamline tool-augmented LLM inference. RL-based approaches (Jin et al., 2025b; Chen et al., 2025; Song et al., 2025) further train LLMs to interleave reasoning and search without supervised trajectories. This end-to-end paradigm internalizes agent behavior and reduces the engineering overhead of hand-designed multi-agent systems, but it remains constrained by instability and performance saturation in open-ended search domains. DSPO directly targets these bottlenecks by providing a stable and sample-efficient policy-optimization framework for learning multi-turn search strategies.

3 METHODOLOGY

This section formalizes agentic search as an RL problem, reviews the relevant policy-gradient objectives, and identifies the failure modes that arise in this setting. We then specify **Dynamic-filter Sequence-level Policy Optimization** (DSPO), including the mechanisms responsible for stable updates and efficient use of sparse rewards, before presenting the complete training procedure.

3.1 PRELIMINARIES

Policy Gradient Methods for LLMs. RL training for LLMs commonly uses policy-gradient algorithms such as PPO (Schulman et al., 2017), which is widely used in LLM alignment. PPO updates a policy π_θ by maximizing a clipped surrogate objective estimated from samples generated by a stale policy $\pi_{\theta_{\text{old}}}$. With token-level averaging, the objective is:

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t|x, y_{<t})}$ denotes the token-level importance ratio. PPO also requires a separately trained value model to estimate token-level advantages $\hat{A}_t$ via Generalized Advantage Estimation (GAE) (Schulman et al., 2015); this auxiliary model increases memory consumption and can introduce additional instability.

GRPO (Shao et al., 2024) removes this value-model dependency by sampling a group of G responses $\{y_i\}_{i=1}^G$ for a prompt x and estimating each response advantage by normalizing its reward relative to group-level statistics. Despite this sequence-level advantage estimate, the optimization objective remains token-level:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) - \beta D_{KL}(\pi_\theta || \pi_{\text{ref}}) \right], \quad (2)$$

where $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}$. For all tokens $y_{i,t}$ within response y_i , GRPO assigns the same sequence-derived advantage:

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}, \quad (3)$$

Thus, every token in response y_i is weighted by the identical advantage $\hat{A}_i$, even though the importance ratio remains token specific.

Agentic Search as a Markov Decision Process. We represent iterative agentic search and reasoning as a discrete-time, finite-horizon Markov Decision Process (MDP).

- **State (s_t):** At turn t , state s_t encodes the complete interaction history: the initial question q , all previously generated reasoning segments and search queries, and all evidence returned by the environment. Since

retrieved evidence must be interpreted and incorporated into later decisions, the state expands with the trajectory and induces long-horizon dependencies that are difficult for conventional token-level RL objectives to optimize.

- **Action** (a_t): An action is a complete textual segment generated by π_θ , consisting of free-form reasoning followed by a control decision. The segment terminates either with `</tool_call>`, which invokes the search engine and inserts returned results within `</tool_response>`, or with `</answer>`, which terminates the trajectory. This structured action space requires the policy to learn both *what* content to generate and *when* to query external information, producing substantial variability in trajectory length.
- **Policy** (π_θ): The policy π_θ is the underlying LLM and generates tokens autoregressively conditioned on s_t . Although the policy conditions on the full interaction history, the optimization loss is computed only over model-generated reasoning and action tokens; environment-returned retrieval content is masked out (Jin et al., 2025b). Optimizing this policy therefore requires credit assignment over long sequences whose intermediate reasoning steps receive no direct supervision or intermediate reward.
- **Trajectory** (τ): A trajectory $\tau = (s_1, a_1, \dots, s_T, a_T)$ contains the full sequence of reasoning segments, tool calls, and environment responses for a question. Since the terminal reward is assigned to the complete trajectory, the optimization target is inherently sequence-level: early query and reasoning decisions can causally affect final-answer correctness.
- **Reward** ($R(\tau)$): We use a sparse terminal reward. A trajectory receives $R = 1$ when the final answer contains the ground-truth text and $R = 0$ otherwise:

$$R(\tau) = \begin{cases} 1 & \text{if } a_{\text{gold}} \subseteq a_{\text{pred}}, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

RL with a Search Engine. Following Jin et al. (2025b), we include the search engine $\mathcal{S}$ as an explicit component of the environment. The LLM policy π_θ generates trajectories by alternating between reasoning tokens and calls to $\mathcal{S}$. The training objective is to maximize expected terminal reward while applying KL regularization that constrains deviations from a reference policy π_{ref} :

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x; \mathcal{S})} [R(x, y)] - \beta D_{\text{KL}} [\pi_\theta(y|x; \mathcal{S}) || \pi_{\text{ref}}(y|x; \mathcal{S})]. \quad (5)$$

Here, $y \sim \pi_\theta(\cdot|x; \mathcal{S})$ indicates that trajectory generation is a multi-step process involving both autoregressive policy outputs and observations returned by the search engine $\mathcal{S}$.

Motivation for DSPO. Although frameworks such as Search-R1 (Jin et al., 2025b) formulate agentic search as an RL problem, direct application of PPO or GRPO remains problematic. The open-ended search environment amplifies the variance of token-level optimization, especially because reward assignment occurs at the sequence level while likelihood-ratio correction is applied per token (Zheng et al., 2025). This mismatch produces noisy gradient estimates over long trajectories and can result in policy collapse. In addition, the sparse binary reward often yields batches containing only successful trajectories or only unsuccessful trajectories, which produces degenerate advantages and eliminates the effective learning signal (Yu et al., 2025; Liu et al., 2025). DSPO is designed to address both failure modes explicitly.

3.2 DYNAMIC-FILTER SEQUENCE-LEVEL POLICY OPTIMIZATION

DSPO modifies prior policy-optimization methods in two ways: (1) it performs likelihood-ratio correction and clipping at the sequence level, thereby aligning the update with trajectory-level rewards; and (2) it applies dynamic filtering so each retained training group contains non-zero reward variance and therefore a non-degenerate advantage estimate. Figure 1 summarizes the resulting training process.

3.2.1 SEQUENCE-LEVEL POLICY OPTIMIZATION FOR ENHANCED STABILITY

Following the sequence-level formulation of GSPO (Zheng et al., 2025), DSPO replaces token-specific importance ratios with a response-level ratio. For response y_i , the sequence-level importance ratio $s_i(\theta)$ is defined as the geometric mean of the constituent token-level ratios:

$$s_i(\theta) = \left(\frac{\pi_\theta(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})} \right). \quad (6)$$

The exponent implements length normalization, which reduces sensitivity to response length and keeps $s_i(\theta)$ on a comparable numerical scale across trajectories. This scale control is essential for stable clipping in long sequences.

Gradient Analysis. The following comparison characterizes the stability benefit of DSPO. In the unclipped token-level GRPO objective, each token log-probability gradient is multiplied by a potentially noisy token-specific weight $r_{i,t}(\theta)$. In the DSPO objective, the log-probability gradients over the sequence are instead scaled by one response-level weight $s_i(\theta)$:

$$\nabla_{\theta} J_{\text{GRPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot \sum_{t=1}^{|y_i|} r_{i,t}(\theta) \nabla_{\theta} \log \pi_{\theta}(y_{i,t} | \dots) \right] \quad (7)$$

$$\nabla_{\theta} J_{\text{DSPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot s_i(\theta) \cdot \sum_{t=1}^{|y_i|} \nabla_{\theta} \log \pi_{\theta}(y_{i,t} | \dots) \right] \quad (8)$$

Because a single correction factor is applied to the entire trajectory, DSPO avoids accumulating independent token-level ratio noise across long responses. The dynamic filtering mechanism complements this sequence-level correction by constructing batches from rollout groups that include both successes and failures, which prevents sparse-reward samples from yielding degenerate advantages $\hat{A}_i$.

3.2.2 DYNAMIC OUTCOME-BASED FILTERING FOR EFFICIENT LEARNING

The binary terminal reward creates a specific failure case for group-based advantage estimation. If all G responses in a group are correct ($R = 1$) or all are incorrect ($R = 0$), the empirical reward variance is zero; consequently, the normalized advantage $\hat{A}_i$ is zero or undefined. These homogeneous groups consume rollout computation but contribute no useful gradient information.

DSPO addresses this issue with a dynamic filtering rule inspired by DAPO (Yu et al., 2025). During sampling, the training buffer retains only trajectory groups containing at least one successful and at least one unsuccessful outcome. Formally, group $\{y_i\}_{i=1}^G$ is eligible for training only when its rewards $\{R_i\}_{i=1}^G$ satisfy:

$$0 < \sum_{i=1}^G R_i < G. \quad (9)$$

This condition enforces non-zero within-group reward variance for every retained group, so group-relative normalization yields a meaningful advantage signal. The dynamic selection procedure therefore converts sparse terminal feedback into a denser and more sample-efficient set of policy updates.

3.3 THE DSPO OBJECTIVE AND TRAINING ALGORITHM

Combining sequence-level ratios with outcome-based filtering yields the DSPO objective. For each valid group sampled from $\mathcal{D}_{\text{filtered}}$, the response advantage $\hat{A}_i$ is computed by group-relative normalization:

$$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R) + \delta}, \quad (10)$$

where δ is a small numerical-stability constant. The policy π_{θ} is updated by maximizing:

$$J_{\text{DSPO}}(\theta) = \mathbb{E}_{(x, \{y_i\}) \in \mathcal{D}_{\text{filtered}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (11)$$

where s_i is the sequence-level importance ratio defined as the geometric mean of token-level ratios. During likelihood computation, all tokens retrieved from the search tool are masked following Jin et al. (2025b); consequently, the objective trains the model to use external evidence in its reasoning and actions rather than to reproduce retrieved text. Algorithm 1 specifies the full training procedure.

4 EXPERIMENTS

This section evaluates Dynamic-filter Sequence-level Policy Optimization (DSPO) along three dimensions: (1) end-task accuracy on challenging question-answering benchmarks; (2) optimization stability relative to baseline methods that exhibit collapse; and (3) the contribution of each algorithmic component, as measured by ablation variants.

Algorithm 1 Dynamic-filter Sequence-level Policy Optimization (DSPO)

```

1: Input: Initial policy  $\pi_{\theta_0}$ , fixed reference policy  $\pi_{\text{ref}}$ , prompt dataset  $\mathcal{D}$ , group size  $G$ , batch size  $B$ , search tool  $\mathcal{R}$ .
2: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_0}$ .
3: for each training step do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$ .
5:   Initialize training buffer  $\mathcal{B} \leftarrow \emptyset$ .
6:   while  $|\mathcal{B}| < B$  do
7:     Sample a prompt  $x \sim \mathcal{D}$ .
8:     Generate a group of  $G$  trajectories  $\{y_i\}_{i=1}^G$  using  $\pi_{\theta_{\text{old}}}$  and the search tool  $\mathcal{R}$ .
9:     Compute terminal rewards  $\{R_i\}_{i=1}^G = \{\text{ContainsAnswer}(y_i, y_{\text{gold}})\}_{i=1}^G$ .
10:    if  $0 < \sum_{i=1}^G R_i < G$  then ▷ Dynamic outcome-based filtering
11:      Add  $(x, \{y_i\}_{i=1}^G, \{R_i\}_{i=1}^G)$  to  $\mathcal{B}$ .
12:    end if
13:  end while
14:  for each  $(x, \{y_i\}, \{R_i\})$  in  $\mathcal{B}$  do
15:    Compute advantages  $\{\hat{A}_i\}_{i=1}^G$  via group normalization of  $\{R_i\}$ .
16:    Compute sequence-level importance ratios  $\{s_i(\theta)\}_{i=1}^G$  using Eq. 6.
17:    Compute the DSPO loss for the group using Eq. 11, applying masks to retrieved tokens.
18:  end for
19:  Update policy parameters  $\theta$  by taking a gradient step on the total loss from  $\mathcal{B}$ .
20: end for

```



Figure 2: **Validation performance of DSPO across seven benchmarks during training.** Accuracy increases steadily across training, indicating that improvements in the RL reward correspond to improved benchmark generalization and that the learned policy supports robust search-and-reasoning behavior.

4.1 EXPERIMENTAL SETUP

Following Search-R1 (Jin et al., 2025b), we use the prompt template in Table 1 to specify the action protocol for the search task. The template defines separate spans for reasoning, tool invocation, and final answering through `<think>`, `<tool_call>`, and `<answer>` markers.

Benchmarks and Baselines. The evaluation protocol follows Search-R1 (Jin et al., 2025b). Training uses a composite dataset formed from the training splits of Natural Questions (NQ) (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018). Generalization is measured on seven QA test sets that cover single-hop, multi-hop, and long-tail factual questions: NQ, TriviaQA (Joshi et al., 2017), PopQA (Mallen et al., 2022), HotpotQA, 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022), and Bamboogle (Press et al., 2022).

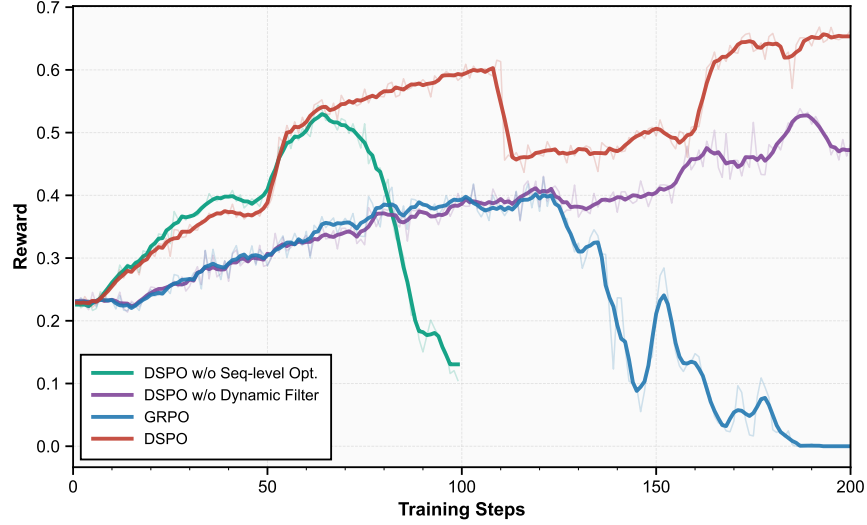


Figure 3: **Training reward dynamics of DSPO and its ablations.** The curves compare optimization behavior across objectives. DSPO (red) improves monotonically and remains stable; token-level variants (green, blue) undergo catastrophic policy collapse; and the sequence-level variant without dynamic filtering (purple) remains stable but converges to a suboptimal plateau.

Prompt Template. Answer the given question. You must conduct reasoning inside `<think>` and `</think>`. first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<tool_call>` query `</tool_call>` and it will return the top searched results between `<tool_response>` and `</tool_response>`. You can search as many times as your want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>`, without detailed illustrations. For example, `<answer>` Beijing `</answer>`. **Question:** ...

Table 1: Prompt template used to constrain model outputs during search-based QA experiments.

The comparison set contains both external baselines and internal ablations. External baselines are Qwen2.5-7B and Qwen2.5-14B models trained with PPO and GRPO under the Search-R1 framework (Jin et al., 2025b;a). To isolate DSPO’s mechanisms, we evaluate two ablated variants: (1) **DSPO w/o dynamic filter**, corresponding to GSPO (Zheng et al., 2025), and (2) **DSPO w/o sequence-level opt.**, which uses the token-level DAPO-style objective (Yu et al., 2025).

For implementation, all 7B experiments initialize from Qwen2.5-7B-Instruct. The training pipeline is built on VeRL (Sheng et al., 2025), with the provided Search-R1-like example code and scripts adapted to implement our objective, filtering rule, and evaluation protocol. Because our experiments use an adjusted reward definition, the PPO and GRPO Search-R1 baselines (Jin et al., 2025b;a) are retrained under the same experimental conditions; these retrained models serve as the primary external comparison points.

Implementation and Evaluation. To isolate the effect of the optimization algorithm, every RL experiment uses the same standard BM25 retriever. This control fixes retrieval capacity across methods, so performance differences primarily reflect the learned search policy and training objective. All methods are trained with a sparse binary reward computed from substring Exact Match (subEM) on the final answer, and subEM is also used as the primary evaluation metric.

4.2 MAIN RESULTS AND ABLATION STUDY

Table 2 reports the main benchmark results together with the ablation study. Presenting these results in a single table makes the interaction between DSPO’s components explicit: the full method can be compared directly with external baselines and with variants that remove either sequence-level optimization or dynamic filtering.

Table 2: Comparison of DSPO, Search-R1 baselines, and DSPO ablations on seven QA benchmarks. Search-R1 baselines include 7B and 14B models trained with GRPO and PPO (Jin et al., 2025b;a). Ablations remove either dynamic filtering (‘w/o dynamic filter’) or sequence-level optimization (‘w/o seq-level opt.’). Parenthesized values are original EM scores from Search-R1; non-parenthesized baseline values are retrained and evaluated under our adjusted reward protocol. Best results are in **bold**; second-best results are underlined.

Dataset	Search-R1				DSPO & Ablations (Ours, 7B)		
	GRPO (7B)	PPO (7B)	GRPO (14B)	PPO (14B)	w/o dyn. filter	w/o seq-lvl opt.	DSPO
NQ	0.423 (0.429)	(0.393)	<u>0.535</u> (0.482)	(0.424)	0.363	0.470	0.580
TriviaQA	0.658 (0.623)	(0.610)	0.760 (0.667)	(0.660)	0.515	0.695	<u>0.754</u>
PopQA	0.395 (0.427)	(0.397)	<u>0.477</u> (0.434)	(0.442)	0.277	0.430	0.498
HotpotQA	0.401 (0.386)	(0.370)	<u>0.563</u> (0.429)	(0.436)	0.330	0.438	0.613
2WikiMultiHopQA	0.357 (0.414)	(0.346)	0.611 (0.424)	(0.379)	0.285	0.398	<u>0.569</u>
Musique	0.122 (0.162)	(0.146)	<u>0.260</u> (0.191)	(0.210)	0.105	0.133	0.270
Bamboogle	0.280 (0.400)	(0.368)	0.504 (0.492)	(0.480)	0.288	0.280	<u>0.432</u>
Average	0.377 (0.396)	(0.385)	<u>0.530</u> (0.446)	(0.433)	0.313	0.406	0.531

Comparison with Baselines. Table 2 shows that the DSPO-trained 7B agent obtains an average score of 0.531. This corresponds to a 34.1% relative improvement over the same-sized Search-R1 GRPO 7B model. The 7B DSPO model also slightly exceeds the average score of the larger Search-R1 14B models trained with either GRPO or PPO. Because retrieval capacity and evaluation conditions are controlled, this result indicates that the gains are primarily associated with the optimization algorithm rather than with increased parameter count.

Analysis of Ablations. The ablation results in Table 2 demonstrate that both DSPO components are necessary for the full effect. Removing the dynamic filter (‘w/o dynamic filter’) reduces the average score to 0.313, indicating that sequence-level optimization alone receives insufficient signal when homogeneous-reward groups dominate sampling. Removing sequence-level optimization (‘w/o sequence-level opt.’) yields an average score of 0.406; although this token-level variant performs better than the filter-free variant, it remains substantially below full DSPO and, as analyzed next, is vulnerable to collapse. The results therefore support the intended decomposition: sequence-level updates provide optimization stability, while dynamic filtering converts sparse rewards into effective group-relative learning signals.

Beyond aggregate metrics, DSPO produces qualitatively structured search behavior, including recognition of irrelevant retrievals, query reformulation, and multi-turn verification (see Appendix A for representative trajectories). These behaviors emerge from RL training under DSPO without supervised search demonstrations.

4.3 ANALYSIS OF TRAINING DYNAMICS

To evaluate optimization stability and sample efficiency, we analyze training reward trajectories for DSPO, its ablations, and key baselines. Figure 3 visualizes these dynamics and isolates the effect of each design choice.

DSPO (red) follows a smooth monotonic trajectory and converges to the highest observed reward level, consistent with stable sequence-level updates. In contrast, the **token-level methods**—DSPO w/o Seq-level Opt. (green) and vanilla GRPO (blue)—show early reward increases followed by catastrophic policy collapse, which is consistent with high-variance token-level gradient corrections. **DSPO w/o Dynamic Filter** (purple) remains stable because it retains sequence-level optimization, but it converges to a lower plateau because homogeneous-reward groups reduce the density of useful updates. These dynamics indicate that stability and efficiency arise from the combination of the sequence-level objective and outcome-based filtering.

To distinguish genuine generalization from reward overfitting, we also monitor validation accuracy throughout training. Figure 2 shows that DSPO accuracy on NQ, HotpotQA, and the other benchmarks increases consistently and tracks the stable reward curve. This correspondence indicates that reward improvements are aligned with the acquisition of a transferable search-and-reasoning policy.

4.4 SCALABILITY AND GENERALIZATION ANALYSIS

We further evaluate whether DSPO transfers across model scale and task domain.

Scalability to Larger Models. We test scalability by applying DSPO to Qwen2.5-14B-Instruct. As reported in Table 3, the DSPO-trained 14B model achieves an average accuracy of 60.6%, compared with 53.0% for the GRPO-14B baseline, giving a relative gain of **14.3%**. The result indicates that the stability benefits of DSPO persist at larger parameter scale and that the method can exploit additional model capacity.

Table 3: Scalability evaluation on Qwen2.5-14B-Instruct. Best results are in **bold**.

Dataset	Instruct (14B)	GRPO (14B)	DSPO (7B)	DSPO (14B)	Gain
NQ	0.345	0.535	0.580	0.629	+17.6%
HotpotQA	0.407	0.563	0.613	0.665	+18.1%
2WikiMQA	0.332	0.611	0.569	0.699	+14.4%
Bamboogle	0.328	0.504	0.432	0.544	+7.9%
PopQA	0.364	0.477	0.498	0.545	+14.3%
TriviaQA	0.643	0.760	0.754	0.802	+5.5%
Musique	0.151	0.260	0.270	0.361	+38.8%
Average	0.367	0.530	0.531	0.606	+14.3%

Generalization to Mathematical Reasoning. We also evaluate whether DSPO generalizes beyond search-based QA by applying it to single-turn mathematical reasoning with models from the Qwen2.5 and Qwen3 families. Table 4 reports results on Math500 and Olympiad-Bench. DSPO exceeds GRPO for both the 7B and 4B settings, suggesting that the same optimization principles are applicable to general reasoning tasks.

Table 4: Generalization evaluation on mathematical reasoning benchmarks. Best results are in **bold**.

Model	Benchmark	Steps	GRPO	DSPO	Gain
Qwen2.5-Math-7B	Math500	200	0.772	0.798	+2.6%
Qwen3-4B	Olympiad-Bench	100	0.728	0.755	+2.7%

5 CONCLUSION

This work addresses two central difficulties in RL training for autonomous LLM search agents: unstable token-level optimization and inefficient learning from sparse terminal rewards. We introduced Dynamic-filter Sequence-level Policy Optimization (DSPO), which combines sequence-level likelihood-ratio correction with dynamic outcome-based filtering. The sequence-level objective reduces the risk of catastrophic policy collapse, while the filtering rule ensures that retained rollout groups contain the reward variance required for informative group-relative advantages. Experiments across challenging QA benchmarks show that DSPO improves task performance and yields more stable training dynamics than prior methods.

Because DSPO learns from environmental feedback alone, it provides a practical route to training search-capable LLM agents without expert demonstration data. The same framework can be extended by substituting stronger retrievers, adding additional tools, or applying the objective to other long-horizon reasoning domains. Since sparse rewards and unstable policy gradients also occur in mathematics and code generation, we expect the two DSPO principles—matching the optimization unit to the reward unit and enforcing signal density—to remain useful beyond agentic search.

REFERENCES

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

- Tuhin Chakrabarty, Vishakh Padmakumar, Faeze Brahman, and Smaranda Muresan. Creativity support in the age of large language models: An empirical study involving professional writers. In *Proceedings of the 16th Conference on Creativity & Cognition*, pp. 132–155, 2024.
- Mingyang Chen, Tianpeng Li, Haoze Sun, Yijie Zhou, Chenzheng Zhu, Haofen Wang, Jeff Z Pan, Wen Zhang, Huajun Chen, Fan Yang, et al. Learning to reason with search for llms via reinforcement learning. *arXiv preprint arXiv:2503.19470*, 2025.
- Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30, 2017.
- Tianzhe Chu, Yuexiang Zhai, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V Le, Sergey Levine, and Yi Ma. Sft memorizes, rl generalizes: A comparative study of foundation model post-training. *arXiv preprint arXiv:2501.17161*, 2025.
- Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, et al. The entropy mechanism of reinforcement learning for reasoning language models. *arXiv preprint arXiv:2505.22617*, 2025.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yixin Dai, Jiawei Sun, Haofen Wang, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2(1), 2023.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. *arXiv preprint arXiv:2011.01060*, 2020.
- Bowen Jin, Jinsung Yoon, Priyanka Kargupta, Sercan O Arik, and Jiawei Han. An empirical study on reinforcement learning for reasoning-search interleaved llm agents. *arXiv preprint arXiv:2505.15117*, 2025a.
- Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. Search-rl: Training llms to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516*, 2025b.
- Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. *arXiv preprint arXiv:1705.03551*, 2017.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick SH Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *EMNLP (1)*, pp. 6769–6781, 2020.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474, 2020.
- Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding rl-zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Hannaneh Hajishirzi, and Daniel Khashabi. When not to trust language models: Investigating effectiveness and limitations of parametric and non-parametric memories. *arXiv preprint arXiv:2212.10511*, 7, 2022.
- Guillermo Marco, Julio Gonzalo, Ramón del Castillo, and María Teresa Mateo Girona. Pron vs prompt: Can large language models already challenge a world-class fiction author at creative text writing? *arXiv preprint arXiv:2407.01119*, 2024.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A Smith, and Mike Lewis. Measuring and narrowing the compositionality gap in language models. *arXiv preprint arXiv:2210.03350*, 2022.

- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems*, 36:53728–53741, 2023.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36:68539–68551, 2023.
- John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. *arXiv preprint arXiv:1506.02438*, 2015.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. In *Proceedings of the Twentieth European Conference on Computer Systems*, pp. 1279–1297, 2025.
- Huatong Song, Jinhao Jiang, Yingqian Min, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, Lei Fang, and Ji-Rong Wen. R1-searcher: Incentivizing the search capability in llms via reinforcement learning. *arXiv preprint arXiv:2503.05592*, 2025.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Trieu H Trinh, Yuhuai Wu, Quoc V Le, He He, and Thang Luong. Solving olympiad geometry without human demonstrations. *Nature*, 625(7995):476–482, 2024.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Musique: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022.
- Junde Wu, Jiayuan Zhu, Yuyuan Liu, Min Xu, and Yueming Jin. Agentic reasoning: A streamlined framework for enhancing llm reasoning with agentic tools. *arXiv preprint arXiv:2502.04644*, 2025.
- Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, et al. Autogen: Enabling next-gen llm applications via multi-agent conversations. In *First Conference on Language Modeling*, 2024.
- John Yang, Carlos E Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. Swe-agent: Agent-computer interfaces enable automated software engineering. *Advances in Neural Information Processing Systems*, 37:50528–50652, 2024.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W Cohen, Ruslan Salakhutdinov, and Christopher D Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. *arXiv preprint arXiv:1809.09600*, 2018.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, et al. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 1(2), 2023.
- Yuzhong Zhao, Yue Liu, Junpeng Liu, Jingye Chen, Xun Wu, Yaru Hao, Tengchao Lv, Shaohan Huang, Lei Cui, Qixiang Ye, et al. Geometric-mean policy optimization. *arXiv preprint arXiv:2507.20673*, 2025.

-
- Chujie Zheng, Shixuan Liu, Mingze Li, Xiong-Hui Chen, Bowen Yu, Chang Gao, Kai Dang, Yuqiong Liu, Rui Men, An Yang, et al. Group sequence policy optimization. *arXiv preprint arXiv:2507.18071*, 2025.
- Qinkai Zheng, Xiao Xia, Xu Zou, Yuxiao Dong, Shan Wang, Yufei Xue, Lei Shen, Zihan Wang, Andi Wang, Yang Li, et al. Codegeex: A pre-trained model for code generation with multilingual benchmarking on humaneval-x. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 5673–5684, 2023.

A APPENDIX

Table A.1 presents representative trajectories that illustrate the search strategies learned by the DSPO-trained model. The examples show that the policy can (1) reformulate queries after irrelevant retrievals, (2) verify candidate evidence with follow-up searches, and (3) extract answer information from retrieved passages.

Table A.1: Representative trajectories illustrating learned search-and-reasoning behavior. The model demonstrates (a) detection of irrelevant retrieval results, (b) query reformulation after failed searches, and (c) verification through additional evidence requests.

Example 1: Presidential Succession (<i>Query Refinement after Failure</i>)	
Question	Who is next in line if something happens to the president?
Gold	vice president
Turn 1	<code>search("who is next in line if something happens to the president")</code> → Retrieved: Documents about novels "Something Happened" and "Mosquitoes Buzz" Agent: "The queries were not directly about the line of succession... The relevant information I need is not present in the search results. Let's try searching again with a more specific query."
Turn 2	<code>search("line of succession for the president")</code> → Retrieved: Philippine and US presidential succession information Agent: "Based on the search results... the next person in line would be the Vice President."
Answer	✓ The Vice President
Example 2: University Identification (<i>Information Verification</i>)	
Question	Which American University is situated in Ithaca, New York?
Gold	Cornell University
Turn 1	<code>search(["American University in Ithaca", "University in Ithaca NY"])</code> → Retrieved: "Ithaca is home to Cornell University, an Ivy League school..." Agent: "Ithaca, New York is home to Cornell University, which is an Ivy League school. However... it is highly likely that the university in question is Cornell. Let me confirm this."
Turn 2	<code>search(["Cornell University location", "Cornell University New York"])</code> → Retrieved: Cornell history confirming establishment in Ithaca Agent: "Cornell University is indeed located in Ithaca, New York."
Answer	✓ Cornell University
Example 3: Biographical Information (<i>Direct Extraction</i>)	
Question	What is John Andrew Martin's occupation?
Gold	lawyer, attorney, politician
Turn 1	<code>search(["John Andrew Martin occupation", "John Andrew Martin job"])</code> → Retrieved: "American journalist, attorney, soldier, and politician..." Agent: "John Andrew Martin was an American <u>journalist, attorney, soldier, and politician</u>. He represented Colorado in the U.S. House of Representatives."
Answer	✓ Journalist, attorney, soldier, and politician